Navigating the Algorithmic Gaze: Critical Perspectives on Advanced Imaging and AI in Visual Heritage

Introduction

The contemporary landscape of visual heritage studies is increasingly shaped by computational methodologies, which offer unprecedented capabilities for analysis, documentation, and dissemination. Advanced imaging techniques and artificial intelligence (AI)-driven tools provide new avenues to explore, understand, and preserve cultural artifacts. However, these technologies are not neutral instruments; they actively mediate, and can potentially distort, our engagement with and understanding of the past. This report aims to critically examine the multifaceted relationship between these sophisticated computational approaches and visual heritage, with a particular focus on inherent risks, systemic failure points, and the overarching necessity for epistemological caution in their application.

Drawing upon a framework of ten specialized inquiries—encompassing challenges from data compression and AI-driven interpretation to ontological modeling and ethical praxis—this analysis will traverse the complex terrain where technology meets cultural patrimony. The objective is to foster a deeper, more critical engagement with these powerful tools, moving beyond simplistic narratives of technological solutionism and toward a more nuanced understanding of their implications. A fundamental tension emerges throughout this examination: a persistent conflict between the drive for computational efficiency and analytical power on the one hand, and the imperative to preserve the semantic, aesthetic, and cultural integrity of visual heritage on the other. This tension is not merely a technical hurdle but a conceptual challenge that necessitates the development of new critical frameworks and evaluation paradigms. For instance, techniques like Principal Component Analysis (PCA) are optimized to capture statistical variance, which may not align with the symbolic importance of subtle visual details.[1] Similarly, AI models, while adept at learning patterns from data [3], can gravitate towards stylistic mimicry rather than genuine iconographic understanding if their training data or learning objectives are not meticulously curated for cultural nuance.[4] These examples underscore a systemic issue requiring more than mere technical refinement; it demands a re-evaluation of how "success," "fidelity," and "understanding" are defined and pursued within the domain of computational heritage.

Section 1: The Double-Edged Sword: Compression, Preprocessing, and the Integrity of Visual Heritage Data

This section addresses the impact of dimensionality reduction and compression techniques, particularly Principal Component Analysis (PCA) applied to multispectral data such as UV-Fluorescence, on the semantic legibility and cultural fidelity of religious iconography. While these methods offer benefits in data management and analysis, their application to culturally rich visual heritage necessitates a critical evaluation of their inherent assumptions and potential for information loss that transcends mere signal degradation.

1.1. PCA in Multispectral Imaging of Artworks: Benefits and Underlying Assumptions

Principal Component Analysis (PCA) is a widely adopted statistical procedure for reducing the dimensionality of large datasets, such as those generated by hyperspectral and multispectral imaging systems. Its primary aim is to transform a set of possibly correlated variables into a smaller set of uncorrelated variables, known as principal components, which capture the maximum variance in the original data, thereby minimizing information loss in a statistical sense.[2] PCA is an adaptive technique, as the principal components are defined by the dataset itself

rather than by a priori assumptions.[2]

In the field of cultural heritage, PCA has demonstrated utility in several applications. When applied to multiband images, including those captured in the ultraviolet (UV) spectrum, PCA can aid in revealing hidden details such as underlying moisture damage, previous restoration efforts, or subtle material differentiations.[7] A case study involving the interior of Saint Irene Chrysovalanto Church, for example, employed PCA on multiband image data to extract such valuable insights for documentation and conservation.[7] Beyond paintings, PCA is also utilized in monument conservation for tasks like classifying constituent materials and investigating the original technology of historic structures. Analysis of bricks from the Hagia Sophia, for instance, used PCA to help determine material provenance and infer aspects of their firing technology.[8] It is often observed that the first few principal components derived from multispectral data contain the most significant portion of the overall variance, and thus, the most "information" from a statistical standpoint.[6] Studies in hyperspectral land use classification have shown, for instance, that using only the first few principal component images can achieve a correct classification rate of around 70 percent.[6]

However, a critical examination of PCA's underlying assumptions is crucial when applying it to iconographic material. The fundamental assumption that maximal statistical variance corresponds directly to maximal semantic or symbolic importance is particularly problematic in the context of religious iconography. Iconography often relies on subtle visual cues, specific color nuances, or minute details to convey complex symbolic meanings. These elements may not necessarily contribute significantly to the overall variance of the multispectral dataset but could be paramount for correct interpretation. Furthermore, PCA assumes that the underlying subspace of the data is linear.[9] The complex, often non-linear relationships between visual features and symbolic meaning in iconography may not be adequately captured by this linear transformation, potentially leading to the misrepresentation or loss of crucial interpretive information.

1.2. Signal Degradation vs. Cultural Fidelity in Compressed UV-Fluorescence Data

Ultraviolet-Fluorescence (UV-F) imaging is an indispensable non-invasive technique in art conservation and technical art history. It exploits the phenomenon where certain materials emit visible light (fluoresce) when irradiated with UV light, allowing for the differentiation of materials on an artwork's surface.[10] Aged varnishes, for instance, often exhibit characteristic green or yellow fluorescence, while more recently applied retouching or restoration materials may absorb UV light and appear dark, enabling conservators to distinguish original layers from later additions.[10] UV-F imaging can also reveal underdrawings, changes in composition (pentimenti), and provide clues about pigment composition based on their fluorescence signatures.[12]

When PCA is applied to UV-F multispectral datasets, typically for data compression or feature enhancement, the process of dimensionality reduction inherently involves a degree of information loss. This loss is quantitatively described by the reconstruction error, which represents the difference between the original data and the data reconstructed from the selected principal components.[15] The reconstruction error can be understood as the variability present in the original data that is not captured by the lower-dimensional representation, or, more formally, as the sum of the eigenvalues corresponding to the discarded (ignored) principal components.[15]

The critical concern in the context of cultural heritage is whether this "signal degradation"—the information loss inherent in PCA-based compression—disproportionately impacts "cultural fidelity." Cultural fidelity, in this sense, refers to the faithfulness of the digital representation to the original artwork's capacity to convey its intended cultural, symbolic, and historical meanings. If the principal components discarded by PCA are those that represent subtle variations in fluorescence—variations that might indicate the use of a specific, symbolically significant

pigment by the original artist, or a particular degradation product that offers clues to the artwork's history or material composition—then cultural fidelity is compromised. This can occur even if the overall "quality" of the compressed image, as measured by standard image quality metrics, remains high. While some advanced PCA methods like Noise-Corrected PCA (NC-PCA) aim to selectively remove noise while retaining structured data [1], the distinction between "noise" and subtle, low-variance "signal" carrying symbolic information is not always clear-cut and depends heavily on the specific context of the artwork.

More generally, lossy compression techniques, by their very definition, permanently remove data from image files. This discarded data can include subtle nuances in color, texture, and depth.[16] While this results in smaller file sizes, the irretrievable loss of information can render the compressed images unsuitable for detailed scholarly analysis or long-term archival purposes, especially if the compression is aggressive.[16] This poses a direct and significant threat to the preservation of cultural fidelity, as the very details that carry meaning might be permanently erased.

A fundamental issue arises from what can be termed the **"Variance-Semantic Gulf"** in PCA for iconography. UV-Fluorescence data often reveals subtle material differences or degradation patterns critical for iconographic interpretation, such as identifying original pigments versus later retouches, or understanding an artist's specific techniques.[7] These critical signals may be characterized by low overall variance within the multispectral dataset, particularly if they are localized to small areas or represent subtle chemical changes. PCA, by its design, prioritizes the capture of high-variance components in the data; the principal components are sorted in decreasing order of variance.[1] Consequently, during compression or dimensionality reduction, lower-variance components are often discarded under the assumption that they represent noise or less significant information.[1] If semantically crucial iconographic details—for example, a faint trace of a specific pigment under UV light indicating an original symbolic color choice, or a subtle change in fluorescence revealing a nearly invisible element of an underdrawing—happen to be low-variance signals, PCA will inherently de-prioritize and potentially discard them. This creates a "gulf" where the statistical measure of importance (variance) diverges significantly from the humanistic or cultural measure of importance (semantic content). The outcome is not merely a generic "information loss" but a targeted loss of culturally significant information. This implies that the standard application of PCA to UV-F iconographic data risks inadvertently erasing the very nuances it is intended to help analyze, necessitating modified PCA approaches or alternative dimensionality reduction techniques that can be guided by semantic priors or expert knowledge.

1.3. Latent Symbolism Retrieval Error and Failure-Informed Thresholding

"Latent symbolism retrieval error" refers to the failure to recover, identify, or correctly interpret underlying symbolic meanings from an artwork after it has undergone data compression or other processing. If the principal components that are crucial for distinguishing specific symbolic elements—or the subtle visual cues that constitute them—are discarded during PCA-based compression, the ability to retrieve these symbols from the processed data is diminished or lost entirely. The assumption that PCA-like mechanisms within more complex models (such as self-attention in transformers) effectively retrieve latent features requires careful empirical validation; for instance, some research suggests a lack of strong empirical support for the idea that self-attention directly implements kernel PCA.[18]

The concept of "failure-informed thresholding," as introduced in Prompt 1, suggests a paradigm shift in how compression thresholds are determined. Instead of relying solely on technical metrics (like allowable reconstruction error or file size targets), this approach implies adjusting compression parameters—such as the number of principal components to retain or the quantization levels in a lossy scheme—based on whether critical semantic information is lost or preserved. This is fundamentally different from standard image thresholding techniques, which are typically based on pixel intensity distributions for segmentation purposes.[19] In the context of

iconography, a thresholding "failure" in compression would occur when a symbol becomes illegible, its distinguishing features are obscured, or its meaning is altered due to data loss. Implementing failure-informed thresholding necessitates an iterative process involving expert validation, where art historians or conservators assess the semantic impact of different compression levels.

The core challenge lies in objectively defining what constitutes "erased meaning." This assessment cannot be based solely on pixel-difference metrics, as highlighted by the constraint in Prompt 3 regarding color transformations. Instead, it requires a semantic evaluation focused on the legibility and interpretability of specific iconographic symbols and attributes.

This leads to an important epistemological shift from considering "loss of detail" to recognizing "loss of meaning" in the context of compressing cultural heritage. Discussions of lossy compression often center on the degradation of visual quality, such as the "loss of detail," "nuance, color, and depth".[16] However, in iconography, these "details" are frequently not merely aesthetic embellishments but are primary carriers of specific symbolic meanings.[20] For example, the precise rendering of a hand gesture, the specific hue of a saint's robe, or the presence of a particular attribute can all be critical for identifying the subject, its theological significance, or its cultural context. Therefore, when compression techniques erase these details, it is not just a technical loss of data points but a potential erasure or alteration of semantic content. A subtle shift in color due to compression might transform a symbol's established meaning entirely. The notion of "failure-informed thresholding" explicitly acknowledges this by proposing that the acceptable limit for compression should be determined by the point at which *meaning* is compromised, not merely when visual artifacts become perceptually noticeable. This re-frames the evaluation of compression algorithms for cultural heritage, moving it from a purely technical or perceptual domain (e.g., assessed by PSNR or SSIM) to a semantic and hermeneutic one, which inherently requires expert human judgment to define and monitor these thresholds of semantic loss.

1.4. Reflexive Trace: Machine-Readable vs. Human-Interpretable Iconography

A critical distinction exists between data that is machine-readable and data that is human-interpretable.[21] Machine-readable data is formatted in such a way that it can be easily processed by a computer, whereas human-interpretable data is that which can be naturally read and understood by humans.[21] Data compression, particularly aggressive or poorly chosen methods, can exacerbate the gap between these two states. An image of a religious icon, after compression, might retain enough coarse features for a machine learning model to classify it broadly (e.g., as "religious art" or "painting"), thus remaining machine-readable for certain tasks. However, the same compression process might have eliminated the subtle details—fine lines, specific color gradations, delicate iconographic attributes—that are essential for a human expert (an art historian, theologian, or conservator) to interpret its specific symbolism, ascertain its date, attribute it to a particular school or artist, or understand its nuanced theological message.

The United States OPEN Government Data Act, for example, defines machine-readable data as "data in a format that can be easily processed by a computer without human intervention while ensuring no semantic meaning is lost".[22] This sets a very high standard, and one that is particularly challenging to meet when compressing complex, information-rich visual heritage like iconography. Human image analysis, as described in the context of medical imaging, involves a process of deconstructing image data and extracting key, salient features that reveal the most important information.[23] If these key features, which in iconography are often subtle and context-dependent, are lost or distorted during compression, then human interpretation is fundamentally impaired, even if the data remains technically processable by a machine.

The critical point of failure, therefore, is reached when a compression algorithm, by prioritizing statistical properties (like variance in PCA) or by aggressively pursuing data reduction targets, strips away the very visual

features that carry nuanced symbolic weight and enable deep human interpretation. At this juncture, the iconographic data might still be "machine-readable" for basic tasks but has become semantically impoverished, rendering it no longer fully "human-interpretable" in its original cultural and symbolic richness. The iconography becomes a dataset that a machine can process but whose deeper meaning is lost to the human expert.

Section 2: AI's Interpretive Lens: Semantic Drift, Hallucination, and Bias in Art Historical Analysis

This section delves into the challenges encountered when Artificial Intelligence (AI) models—including Vision-Language Models (VLPs), iconographic captioning systems, and Convolutional Neural Networks (CNNs) for emotion recognition—are applied to the interpretation of artworks, with a particular focus on religious and stylized iconography. The core issues revolve around semantic inaccuracies, the generation of erroneous or misleading information (hallucinations), and the influence of inherent biases within these computational systems.

2.1. Semantic Drift in Vision-Language Models (VLPs)

Vision-Language Models (VLPs), such as BLIP and Flamingo, are increasingly employed to infer symbolic meanings from visual art, including paintings. However, a significant challenge in this application is "semantic drift," which occurs when the model's interpretation deviates from the intended, historically established, or culturally accurate symbolic meaning of the artwork. A primary failure point arises when the model's understanding of overall "scene consistency" overrides the accurate attribution of specific symbols. For instance, a VLP might correctly identify individual elements within a painting (e.g., "a figure kneeling," "a bright light emanating from above," "expressions of awe or fear on faces") but then synthesize these observations into an interpretation that is visually coherent yet symbolically incorrect. A scene depicting religious ecstasy or a divine visitation might be mislabeled as "a person frightened by an unexpected explosion" if the latter interpretation aligns more strongly with patterns learned during the model's training on general-domain images, even if it contradicts known iconographic conventions for the specific artwork.

This "failure layer" becomes evident when the model's visual grounding—what it appears to "see" in the image—directly contradicts textual heritage, which encompasses the body of scholarly knowledge about iconographic traditions, symbolic codes, and historical context.[20] The symbol grounding problem is central here: LLMs and MLLMs often process symbols as statistical patterns without a deep, contextual, or experiential understanding of their situated meaning.[24] While MLLMs like GPT-4V demonstrate potential in performing aspects of art analysis, such as identifying artistic styles, artists, or describing scene elements, they frequently struggle with the nuanced interpretation of symbolic meanings and the deep historical significance embedded in artworks.[25]

The "meta-loop" involving a historian's critique is therefore indispensable. An art historian, equipped with knowledge of specific iconographic programs, symbolic attributes, and cultural contexts [20], would critically evaluate an AI-generated caption or interpretation. They would likely identify anachronisms (assigning meanings or objects not plausible for the artwork's period), misidentified symbols (e.g., mistaking one saint's attribute for another's), or culturally inappropriate interpretations that arise from the VLP's semantic drift. This expert feedback is crucial for identifying the limitations of current VLPs and guiding the development of models with more robust symbolic grounding.

A core difficulty is the **"Symbolic Grounding Impasse"** in AI art interpretation. AI models, particularly VLPs and MLLMs, learn from vast datasets by identifying statistical correlations between visual patterns and co-occurring textual descriptions.[3] However, iconography and symbolic art rely on culturally and historically situated meanings,

where symbols are not merely visual patterns but referents to complex concepts, narratives, and belief systems.[20] This is the essence of the "symbol grounding problem"—the challenge of connecting abstract symbols to their real-world or conceptual meanings in a way that mirrors human understanding.[24] Current AI often fails to bridge this gap effectively, leading to "semantic drift" where interpretations are visually plausible but symbolically incorrect, or "hallucinations" that represent a fundamental break in symbolic logic or factual accuracy.[27] Models may achieve "scene consistency" or stylistic mimicry [4] because these aspects are often learnable from surface-level visual features. However, this does not equate to an understanding of the underlying symbolic grammar or the specific historical and cultural context that imbues symbols with meaning.[25] This "impasse" becomes particularly evident when the AI's visual grounding contradicts established textual heritage or when it misinterprets ritual scenes based on superficial visual cues rather than their symbolic intent.[30] Consequently, true iconographic understanding by AI necessitates more than pattern matching; it requires the integration of external knowledge graphs, historical contextual information, and potentially causal reasoning about symbolic systems. The historian's critique then becomes an essential mechanism for identifying and attempting to correct these fundamental grounding failures.

2.2. Iconographic Captioning and Hallucination Detection

The task of generating accurate and meaningful captions for iconographic artworks using AI presents unique challenges, particularly concerning the detection and mitigation of "hallucinations." Hallucinations in the context of VL models refer to the generation of textual descriptions that include non-existent objects or attributes, misrepresent relationships between elements, or introduce information that is factually incorrect or not visually grounded in the image.[27] Frameworks like HalCECE (Hallucination Detection through Conceptual Counterfactual Explanations) aim to provide more explainable and fine-grained evaluation of such hallucinations by decomposing concepts within captions and suggesting minimal edits to correct them.[27]

Traditional automated metrics for evaluating image captions, such as BLEU (Bilingual Evaluation Understudy) and ROUGE (Recall-Oriented Understudy for Gisting Evaluation), are largely insufficient for assessing the quality of iconographic captions.[27] These metrics primarily measure n-gram (sequence of n words) overlap between the generated caption and reference captions. While useful for gauging linguistic fluency or similarity in general-domain captioning, they fail to capture semantic accuracy, symbolic correctness, or the nuanced interpretation required for iconographic art. An AI-generated caption could achieve a high BLEU score by using similar vocabulary to a reference caption yet completely misrepresent the symbolic meaning or historical context of the artwork. Consequently, there is a pressing need for alternative evaluation metrics that focus on symbolic misalignment and sensitivity to visual artifacts. Metrics like CAPTURE (CAPtion evaluation by exTracting and coUpling coRE information) [34] or DCScore [38] represent efforts in this direction, aiming to extract and match core visual elements (objects, attributes, relationships) to provide a more reliable assessment of caption quality, particularly for detailed descriptions.

Several analytical lenses are critical for evaluating iconographic captioning systems:

- **Visual-text co-entanglement:** This refers to the intricate and often bidirectional influence between visual features perceived by the model and the textual elements it generates. If an AI model misinterprets a visual cue, it might generate an incorrect textual description. This description, if fed back into a reasoning process or used to guide further attention, can reinforce the initial error, leading to a cascade of inaccuracies. This "co-entanglement" can result in self-reinforcing errors if the model's understanding is not firmly grounded in true iconographic knowledge and relies instead on superficial correlations learned during training.[3]
- **SmoothGrad/Grad-CAM++ explainability flaws:** Explainability techniques like Grad-CAM and its variants (e.g., Grad-CAM++) are used to visualize which parts of an image a model focuses on when making a

prediction or generating a caption.[40] While Grad-CAM++ aims to improve upon Grad-CAM by offering better object localization and handling of multiple instances [42], gradient-based methods, in general, can be brittle, unreliable, and highly sensitive to implementation details, such as the choice of layer from which gradients are extracted.[43] Their explanations may not always be faithful to the model's true underlying reasoning processes.[44] This lack of robustness impacts their utility in reliably guiding the generation of accurate iconographic captions or in diagnosing precisely why a hallucination occurred.

- **Bias drift in mythological vs. secular symbols:** AI models are susceptible to inheriting and amplifying biases present in their training data. This can lead to stereotypical, inaccurate, or culturally insensitive portrayals of religious, mythological, or secular symbols.[47] "Bias drift" refers to how these biases can evolve, become amplified, or subtly shift as models are fine-tuned on new data or applied to different contexts. For instance, reports indicate that models like DALL-E and ChatGPT have struggled to generate common Islamic symbols accurately or have misrepresented them, while readily generating symbols associated with other faiths.[49] Such disparities highlight the unevenness of cultural representation in training datasets and model capabilities.
- **Meta-layer: Learned iconography vs. style mimicry:** This distinction is fundamental. Does the AI model genuinely understand the *meaning* and conventions of iconographic elements (e.g., knowing that a lily symbolizes purity in certain contexts), or is it merely replicating visual styles and co-occurring features that it has learned are associated with particular themes, artists, or periods?.[4] The ability to distinguish authentic human-created art from AI-generated images often hinges on detecting inconsistencies, a lack of deeper contextual understanding, or an over-reliance on superficial stylistic cues in the AI's output.[51] Interestingly, compression artifacts can sometimes make AI-generated images appear *more* realistic by obscuring the subtle digital clues that differentiate them from natural images, potentially masking a lack of true iconographic understanding.[53]

The application of **Explainability Methods (like Grad-CAM++) can be a double-edged sword in iconographic analysis.** While intended to reveal *why* a model makes a certain prediction by highlighting salient image regions [40], their own inherent flaws pose risks. If Grad-CAM++ highlights a region based on stylistic features that the model has merely learned to associate with a certain type of caption (indicative of style mimicry), rather than highlighting a true iconographic symbol whose meaning is understood, then the "explanation" can inadvertently reinforce a superficial understanding. It might even mislead a human user into believing the model comprehends symbols when it does not. Furthermore, if the AI model itself harbors biases—for example, in its interpretation of mythological versus secular symbols [47]—the explainability map will likely reflect this bias, potentially appearing to "justify" a biased interpretation rather than exposing it as a systemic flaw. This underscores that relying on current XAI methods to validate iconographic understanding in AI is a precarious endeavor. The "explanation" provided might be an artifact of the XAI method's own mechanics or simply a reflection of the primary AI model's superficial pattern matching capabilities. Consequently, a critical evaluation of XAI outputs, especially within symbolic and culturally sensitive domains, is as crucial as evaluating the primary AI model's outputs themselves. This reinforces the urgent need for evaluation methodologies that go beyond standard metrics to specifically address "symbolic misalignment" and "artifact sensitivity."

Furthermore, the **cascading effect of artifact sensitivity from image to interpretation** is a significant concern. Image artifacts—whether originating from compression processes (as discussed in Section 1[53]), physical degradation of the original object, or even the image capture process itself—alter the visual data that is fed into AI models. Research indicates that compression can destroy subtle clues that differentiate natural images from AI-generated ones, potentially making AIGIs appear more realistic but also obscuring true features.[53] Similarly, image blur or compression artifacts can introduce noise that directly impacts the accuracy of Optical Character Recognition (OCR) in VLMs.[55] AI models, especially deep learning networks, are known to be sensitive to such

input perturbations. These artifacts can be misinterpreted by the model as actual features or, conversely, can obscure genuine features that are critical for correct interpretation.[54] In the context of iconographic captioning, this "artifact sensitivity" means that the model might generate captions based on these visual artifacts rather than the true content of the artwork, leading to hallucinations or significant symbolic misalignments. For example, a compression artifact could be erroneously interpreted as a non-existent object, detail, or even a symbolic element. In emotion recognition from art, artifacts could alter textures or color relationships in a way that leads to misclassification of the intended emotional content. If an artistic stylization itself is treated as an "artifact" from the perspective of a model predominantly trained on photorealistic images, this can lead to the "stylization-induced semantic void" where the model fails to extract meaningful emotional cues. This creates a detrimental cascade: artifacts in the image data lead to errors in feature extraction by the AI, which in turn lead to flawed interpretations (such as hallucinated captions or incorrect emotion labels), ultimately undermining the utility and reliability of AI in art historical analysis. Studies show that Multimodal Large Language Models (MLLMs) can struggle with visually confusing images [56], although techniques like Retrieval-Augmented Generation (RAG) show promise in reducing hallucinated image artifacts.[57] Therefore, robust preprocessing of images to detect and mitigate artifacts is a crucial preliminary step before applying AI for iconographic interpretation. Moreover, AI models need to be specifically trained or designed to be robust to common types of artifacts found in digitized heritage imagery, or at the very least, to flag uncertainty when such artifacts are detected. The evaluation of "artifact sensitivity" is thus a critical component of assessing model suitability for cultural heritage applications.

2.3. Emotion Recognition in Iconographic Paintings

The application of Convolutional Neural Networks (CNNs) to recognize emotions evoked by artworks, including iconographic paintings, is an emerging area of research.[58] While CNNs have shown success in emotion recognition from facial expressions in photographs, their deployment on stylized or symbolic art presents distinct challenges. These challenges stem from differences in artistic styles, the impact of image preprocessing (such as resizing, which can cause distortion [58]), and common machine learning issues like overfitting, where the model learns the training data too well but fails to generalize to new, unseen artworks.[58] Features such as compositions, color tones, and brush strokes are known to affect human emotional interpretation of paintings [58], but how effectively CNNs can learn these complex relationships in non-photorealistic contexts is an open question.

A specific typology of failures can be anticipated when CNNs attempt emotion inference from stylized or symbolic artworks:

- **Misclassification of ritual scenes as “aggressive”:** Religious iconography often depicts ritual scenes that involve dramatic gestures, intense facial expressions, or dynamic compositions intended to convey spiritual fervor, divine power, or solemnity. If a CNN trained primarily on datasets of everyday human emotional expressions (e.g., happy, sad, angry based on photographs) encounters these iconographic representations, it may interpret them literally, devoid of their symbolic context.[20] For example, a figure with wide eyes and an open mouth, intended to depict awe or divine inspiration in a ritual context, might be misclassified as "fearful" or "surprised" in a mundane sense. Similarly, dynamic portrayals of divine judgment or spiritual struggle might be misinterpreted as "aggressive." The symbolic language that imbues these representations with specific religious meaning is lost on the AI, which lacks the cultural and theological framework for correct interpretation.[31] This aligns with broader arguments that AI, lacking life experience and genuine emotions, cannot truly grasp the deep meaning embedded in art.[30]
- **Stylization-induced semantic void:** Artistic stylization is a hallmark of much iconographic and religious art. This can include non-naturalistic features such as elongated figures, symbolic color palettes (e.g., gold backgrounds representing divinity), hierarchical scaling, and abstracted forms. Such stylization can create a "semantic void" for CNNs trained on photorealistic images. The visual cues that these models have learned to

associate with particular emotions (e.g., subtle muscle movements in a human face, typical body language) may be absent or radically transformed in stylized art. Research has shown that applying artistic filters (e.g., a Van Gogh-esque style extracted via a CNN) to photographs significantly alters the perceived emotional response in human viewers, suggesting that stylization itself is a powerful modulator of emotion.[59] If an AI model cannot parse or account for this stylization—if it attempts to apply emotion recognition features learned from realistic images directly to a highly stylized icon—its ability to infer the intended or perceived emotion will likely fail. The features it seeks are either not present or are used in a completely different, symbolic manner.

- **Drift Layer: Texture dominance erasing emotional intention:** CNNs, particularly in their early layers, are highly sensitive to texture. There is evidence that some CNN architectures exhibit a bias towards texture over shape or more global compositional cues in image classification tasks.[60] In iconographic art, emotion is often conveyed through symbolic means—such as specific postures, gestures, attributes, colors, or the overall narrative context—rather than through realistic depiction of facial muscle textures or subtle skin tones that might indicate emotion in a photograph. If a CNN over-relies on textural features present in the artwork (e.g., the texture of aged varnish, the craquelure of paint, the weave of a canvas, the pattern of gold leaf tooling, or even specific types of brushstrokes) that are not directly correlated with the intended emotional expression within that specific symbolic system, it will lead to misinterpretation. For instance, concepts like "Texture Key Driver Factors" (TKDFs), identified for facial expression recognition from photographs (focusing on micro-changes in skin texture around brows, eyes, and mouth [62]), would be largely irrelevant or misleading if applied to iconographic art where such naturalistic texture cues for emotion are deliberately absent or replaced by symbolic conventions. The AI's focus on irrelevant textures could thus erase or override the actual, symbolically conveyed emotional intention of the artwork.

The following table summarizes some of the key semantic and interpretive challenges discussed:

Table 1: Semantic and Interpretive Challenges in AI-Driven Analysis of Visual Heritage

AI Task	Potential Semantic/Interpretive Error Type	Impact on Cultural/Historical Understanding	Proposed Non-Standard Evaluation Approaches	Relevant References
Symbolic Meaning Inference (VLP)	Semantic Drift (Scene consistency overriding symbolic attribution)	Misrepresentation of artist's intent, anachronistic interpretation, distortion of theological/narrative meaning.	Historian's critique, symbol grounding validation, comparison with textual heritage.	24
Iconographic Captioning	Hallucination (Object, Attribute, Relational, Role), Symbolic Misalignment	Generation of factually incorrect or misleading descriptions, obscuring true iconographic content, creation of false narratives.	Metrics beyond BLEU/ROUGE (e.g., CAPTURE, DCScore), HalCECE framework, artifact sensitivity analysis, symbolic misalignment metrics.	27
Iconographic Captioning	Bias in Symbol Interpretation (e.g., Mythological vs. Secular, inter-religious	Perpetuation of stereotypes, misrepresentation of cultural/religious	Audits for bias in training data and model outputs, comparative analysis across	47

	bias)	significance, inequitable representation of diverse traditions.	different cultural/symbolic systems.	
Iconographic Captioning / Art Analysis	Learned Style Mimicry (vs. True Iconographic Understanding)	Superficial interpretation based on visual style rather than symbolic content, failure to grasp deeper meanings or artist's conceptual framework.	Evaluation of generalization to novel symbolic combinations, testing for understanding of symbolic grammar vs. pattern repetition.	4
Emotion Recognition from Art	Misclassification of Ritual Scenes (e.g., as "aggressive")	Failure to understand symbolic context of expressions/gestures, imposing inappropriate emotional labels based on literal visual cues.	Integration of contextual knowledge, evaluation by experts in religious/ritual studies.	20
Emotion Recognition from Art	Stylization-Induced Semantic Void	Inability to interpret emotion from non-naturalistic representations where standard emotional cues are absent or transformed by artistic style.	Training on diverse artistic styles, evaluating robustness to stylization, models that can differentiate style from emotional content.	59
Emotion Recognition from Art	Texture Dominance Erasing Emotional Intention	Over-reliance on visual textures irrelevant to symbolic emotional cues, leading to misinterpretation of artist's intended emotional expression.	Models that can weigh texture, shape, and symbolic cues appropriately for art; analysis of feature importance.	60

Section 3: The Fidelity Paradox: Technical Accuracy vs. Aesthetic and Semantic Faithfulness in Digital Pipelines

The digitization of visual heritage involves complex pipelines that aim to capture and represent artworks with high technical accuracy. However, a paradox often emerges: processes optimized for technical fidelity may inadvertently compromise the aesthetic and semantic faithfulness of the digital representation. This section explores this tension through three distinct lenses: color transformations, High Dynamic Range (HDR) imaging, and aerial photogrammetry, highlighting how seemingly accurate technical operations can lead to significant losses in perceived meaning and artistic intent.

3.1. Cross-Gamut Mistranslation in Digitized Color Pipelines

The accurate reproduction of color is paramount in the digitization of cultural artworks, as color is often a key carrier of aesthetic, symbolic, and historical information. Digital color pipelines frequently involve transformations between different color spaces, such as from an acquisition space (e.g., camera RGB) to a standard working space (e.g., Adobe RGB), and finally to a display space (e.g., sRGB or YRGB for certain displays) or a perceptually uniform space like CIELAB, which is designed to align with human color perception. While these transformations are mathematically defined to ensure technical fidelity—meaning that color values are converted correctly according

to established formulas—they can lead to "mistranslations" in perceived chroma (color saturation or intensity) and hue when viewed by a human observer.

A critical constraint in assessing these mistranslations is that evaluation must not rely solely on pixel-difference metrics (e.g., ΔE in CIELAB). While such metrics can quantify the numerical difference between an original and a transformed color, they do not fully capture the impact on aesthetic fidelity or symbolic meaning. For instance, a mathematically small color shift might cross a critical perceptual boundary, causing a vibrant, sacred red to appear dull and lifeless, or a subtle, intentionally muted tone to become overly saturated, thereby altering the artwork's mood or the symbolic resonance of the color. The failure stack often occurs where the transformation preserves technical fidelity (e.g., L*, a*, and b* values are correctly calculated from RGB inputs) but fails aesthetic fidelity (the perceived vibrancy, harmony, or emotional impact of the color relationships within the artwork is distorted).

This discrepancy often reflects color biases baked into the imaging tools and color management systems themselves. Many such systems are optimized for photographic realism or for the gamuts of contemporary display technologies. They may not be adequately equipped to handle the specific, often unique, gamuts of historical pigments, the effects of aging on color, or the particular color relationships intended by artists from different periods and cultures. This necessitates a reflexive insight loop, questioning the universality of our color tools and their suitability for the diverse chromatic C_H_A_R_A_C_T_E_R of heritage objects.

The **"Perceptual-Semantic Gap"** in color and tone management is a significant manifestation of this fidelity paradox. Digital color management systems and HDR pipelines are founded on mathematical models of color (e.g., CIELAB, YRGB) and tone (e.g., logarithmic mapping) that strive for technical accuracy and consistency. However, human perception of color and tone is not a purely mathematical process; it is profoundly influenced by context, such as surrounding colors and tones, cultural associations with specific hues, and the artist's expressive intent.[20] "Cross-gamut mistranslation" arises when a technically correct color transformation results in a perceived color that no longer conveys the same aesthetic or symbolic meaning. For example, a vibrant sacred red, crucial for its symbolic power, might appear desaturated or shifted in hue, diminishing its impact. Similarly, subtle shadow details, meticulously rendered by an artist to create a specific mood or to guide the viewer's eye, might be lost or over-emphasized in HDR tone mapping, even if local tonal values are "accurately" represented. This gap exists because computational models typically optimize for the physical properties of light and color or for the capabilities of display devices, whereas artistic interpretation and appreciation rely on a complex interplay of these physical properties with human psychology, cultural conditioning, and learned visual conventions. Addressing this gap requires digital pipelines for visual heritage that incorporate perceptual and semantic considerations, not just technical ones. This might involve developing perceptually-aware color mapping algorithms, tone mapping strategies that are sensitive to artistic intent (perhaps guided by computational style analysis), and evaluation metrics that extend beyond simple pixel differences to assess aesthetic and symbolic fidelity. The inherent biases within existing tools, often designed with contemporary photography in mind, call for a deconstruction and re-evaluation of their foundational assumptions when applied to the unique C_H_A_R_A_C_T_E_R_I_S_T_I_C_S of historical artworks.

3.2. Tonality Thresholds as Aesthetic Agents in HDR Pipelines

High Dynamic Range (HDR) imaging aims to capture and represent a much wider range of luminance values than traditional imaging, from the deepest shadows to the brightest highlights, more closely mimicking the capabilities of human vision. When applied to cultural heritage, particularly paintings or architectural interiors with complex lighting, HDR can reveal details that would be lost in standard dynamic range images. However, to display HDR content on conventional screens, a process called tone mapping is required to compress this wide dynamic range. Adaptive logarithmic HDR systems often employ sophisticated tone mapping operators, which may use techniques like layer masks whose opacity is controlled by tonal thresholds, effectively applying different adjustments to different brightness regions of the image.

The critical issue arises when these "tonal thresholds" in layer masks begin to act as "aesthetic agents," interfering with the intended visual mood, painterly qualities, or perceived depth of the artwork. A failure reflection occurs when computational tone accuracy conflicts with perceived artistic integrity. For example, while local tonal adjustments might accurately represent the brightness levels in specific areas of a painting (computational tone accuracy), these adjustments can disrupt the overall tonal harmony and the subtle relationships between light and shadow meticulously crafted by the artist. This can lead to a flattening of the image, a loss of atmospheric perspective, or what is termed "perceived depth collapse." The nuanced interplay of light and shadow that creates volume, defines form, and evokes a particular mood can be inadvertently destroyed.

Furthermore, "layer mask opacity ambiguity" can act as a "hallucination vector." If the algorithms or rules determining the opacity of these masks are not perfectly aligned with the image content or the artist's original intent, they can introduce unnatural tonal transitions, create artificial-looking halos around objects, or selectively highlight or suppress details in a way that distorts the image. These effects can be considered visual "hallucinations"—artifacts that were not present in the original artwork and that misrepresent its appearance.

A significant meta-loop question is whether such a system can detect its own tonal "aesthetic inversion"—that is, can it recognize when its technically "correct" adjustments have resulted in an aesthetically undesirable or misleading outcome? This implies a need for systems to evaluate their output not just for technical correctness (e.g., ensuring all details are visible within the display's dynamic range) but for aesthetic coherence or faithfulness to an artistic style or intent. This is an exceptionally challenging task, likely requiring the integration of art historical knowledge or style-based analysis into the HDR processing pipeline.

3.3. Dynamic Range Failure in Optical Flow for Aerial Photogrammetry

Aerial photogrammetry is increasingly used for the 3D reconstruction of heritage sites, providing valuable models for documentation, analysis, and conservation planning. This technique relies on capturing multiple overlapping images, often from drones or aircraft, and then using algorithms like Structure-from-Motion (SfM) to reconstruct the 3D geometry. A key component of SfM is optical flow, which tracks corresponding feature points across these multiple images. When HDR footage is used for aerial photogrammetry, particularly for sites with extreme lighting conditions (e.g., bright sunlit facades adjacent to deep, shadowed courtyards), the dynamic range of the scene can exceed the compensation ability of optical flow algorithms.

The task is to map this "failure boundary." High luminance contrast poses a significant challenge because feature descriptors can become unreliable. A feature point that is well-illuminated and rich in detail in one image might be plunged into deep shadow (and thus exhibit low contrast and few discernible details) in an overlapping image taken from a slightly different viewpoint or at a slightly different time. This makes robust matching of feature points across images difficult, leading to errors in the optical flow estimation and, consequently, inaccuracies or failures in the 3D reconstruction. A crucial constraint in investigating this problem is the inclusion of a simulation of information degradation under high luminance contrast. This involves modeling how feature distinctiveness diminishes, how matching scores degrade, and how tracking ultimately fails as the contrast between corresponding regions in different images becomes too extreme.

The meta-insight emerging from this challenge concerns the limits of SfM in such conditions: when should SfM, which is heavily reliant on robust optical flow, be abandoned or supplemented? The proposed alternative is a move towards "depth + tone fusion." This could involve integrating depth data from other sensors (like LiDAR) with the tonal information from HDR images. In such a hybrid approach, the precise geometric information from LiDAR could provide a robust scaffold for the 3D model, while the rich tonal information from the HDR imagery could be used to refine surface details, texture the model, or even guide the interpretation of ambiguous depth readings, rather than relying solely on feature matching across extreme dynamic ranges that push optical flow algorithms beyond their limits.

The concept of a **"failure boundary" as a catalyst for hybrid methodologies** is particularly pertinent here. The failure boundary of optical flow in HDR photogrammetry due to extreme dynamic range (Prompt 9) illustrates a point where a specific computational technique demonstrably breaks down. Relying solely on that technique beyond this boundary leads to significant information degradation and unreliable results, such as failed or grossly inaccurate 3D reconstructions. The meta-insight—questioning when SfM should be abandoned in favor of depth + tone fusion—points to a crucial strategic response: recognizing the inherent limitations of one approach and proactively integrating complementary methods to overcome these weaknesses. This is not merely about switching from one technique to another but about intelligently *fusing* information from different sources or modalities (e.g., geometric data from LiDAR, tonal information from HDR images) to create a more robust and accurate outcome than either method could achieve alone. This principle of a "failure boundary" prompting a shift towards hybrid methodologies can be generalized across many areas of computational heritage. For instance, if PCA consistently fails to preserve essential semantic detail in iconographic analysis (as discussed in Section 1), a hybrid approach that combines PCA with feature selection guided by prior semantic segmentation or expert annotation might prove more effective. Similarly, if AI captioning systems persistently hallucinate when describing certain types of iconographic content (as explored in Section 2), a hybrid system that integrates these AI captions with rule-based checks against a curated iconographic knowledge base might be necessary to improve reliability. Thus, research in computational heritage should focus not only on optimizing individual algorithms in isolation but also on rigorously understanding their respective failure boundaries and on developing sophisticated, adaptive hybrid methodologies. Such systems would be capable of dynamically combining techniques to effectively handle the inherent complexities and variabilities of diverse heritage data, requiring a deep, empirical understanding of the error characteristics and operational limits of each constituent method.

Section 4: From Pixels to Meaning: Ontological Modeling and Knowledge Representation for Visual Archives

The transformation of visual heritage from raster images into structured, machine-readable knowledge is a central challenge in digital humanities and archival science. This section addresses the complexities of mapping high-resolution digitized art to semantic formats like RDF (Resource Description Framework) triples, focusing on the promise of ontological interoperability and the significant cognitive and interpretive hurdles involved.

4.1. The Promise and Peril of Semantic Web Technologies for Art

Semantic Web technologies, with RDF at their core, aim to create a web of linked data that is not only human-readable but also machine-interpretable and interoperable.[22] By structuring information into triples—consisting of a subject, a predicate, and an object (e.g., <Mona_Lisa> <created_by> <Leonardo_da_Vinci>)—RDF allows for the formal representation of knowledge and relationships. This holds considerable promise for cultural heritage, potentially enabling the connection of disparate datasets from various institutions, facilitating complex queries across collections, and fostering new research by revealing previously obscured connections between artworks, artists, and cultural contexts. Recent approaches even leverage Large Language Models (LLMs) to assist in creating semantic graphs for navigating cultural heritage datasets, aiming to enhance accessibility and user experience.[65]

However, the "peril" in applying these technologies to art lies significantly in the inherent process of "compression of visual nuance into symbolic triples." Artworks, particularly those rich in iconographic content or complex historical layers, are often characterized by ambiguity, polysemy (multiple meanings), and deep context-dependency. The very act of reducing their multifaceted visual and symbolic information into a series of discrete, atomic triples can lead to oversimplification, the loss of subtle meanings, or the imposition of a singular interpretation where multiple interpretations are valid. The richness of visual experience and the often-elusive

nature of artistic intent can be difficult to capture within the rigid structure of formal ontologies.

This leads to the **"Semantic Bottleneck" of RDF Triplification for Visual Nuance.** Visual artworks, especially complex iconographic pieces, are replete with nuance, ambiguity, and polysemy.[20] A single image can simultaneously convey multiple layers of meaning through its use of color, composition, gesture, texture, and the subtle interplay of its constituent elements. RDF triples, by their fundamental structure of subject-predicate-object, enforce a discrete and relational representation of knowledge.[22] The process of "compressing visual nuance into symbolic triples," as highlighted in Prompt 4, inevitably acts as a "semantic bottleneck." Complex, continuous, allegorical, or ambiguous visual information must be translated into a finite set of predefined predicates and entities within an ontology. For instance, the profound sorrow subtly expressed in the eyes of a painted figure in a religious scene might be reduced in RDF to a triple like <figure_X> <expresses_emotion> <sorrow_concept>. While factually correct at a high level, this triple loses the *how* (the specific visual cues in the eyes, the tilt of the head, the surrounding context) and the *degree* or *quality* of the expressed sorrow. It cannot easily convey whether the sorrow is one of resignation, grief, or empathetic suffering. This bottleneck is further exacerbated by the significant "cognitive load" (Prompt 4) placed on human modelers. These domain experts must make inherently subjective decisions about which nuances are important enough to encode, how to formalize them within the constraints of the chosen ontology, and which aspects must inevitably be omitted, leading to potential information loss or the introduction of bias reflecting the modeler's own interpretive framework. Therefore, while RDF undeniably enhances machine readability and interoperability for certain types of information, it may not be sufficient as the sole method for representing the full semantic richness and interpretive depth of visual heritage. Mitigating this bottleneck likely requires complementary approaches, such as linking RDF statements to rich media annotations (e.g., highlighting specific image regions and associating them with detailed textual descriptions), incorporating qualitative narratives, or developing more expressive ontological models. The inherent "friction between database logic and visual ambiguity" (Prompt 4) remains a persistent and fundamental challenge in this domain.

4.2. Cognitive Load in Ontology Modeling vs. Interpretative Freedom

The development of a comprehensive and accurate ontology for a domain as complex as art history imposes a significant "cognitive load" on the domain experts involved. These experts—art historians, curators, conservators—must meticulously define classes of objects (e.g., "Icon," "Fresco," "Manuscript Illumination"), types of symbols, attributes of artworks (e.g., "material," "creation_date," "style"), and the myriad relationships between them (e.g., "depicts_scene," "created_by_artist," "influenced_by_style," "symbolizes_concept"). This process is not merely technical; it is profoundly interpretive, requiring consensus on definitions and classifications that may themselves be subjects of scholarly debate.

This necessary structuring of knowledge through an ontology can, however, conflict with the principle of "interpretative freedom." An ontology, by its nature, imposes a particular worldview, a specific classificatory scheme, and a set of predefined relationships onto the domain it describes. While this formal structure is what enables computational reasoning, automated querying, and data interoperability, it may not easily accommodate alternative, minority, evolving, or even contradictory interpretations of artworks. The "friction between database logic and visual ambiguity" is central to this tension. Database logic often prefers clear, unambiguous classifications and relationships, while visual art thrives on ambiguity, multiple layers of meaning, and the capacity to evoke diverse responses and interpretations over time.

4.3. Failure Typology in Semantic Mapping

When mapping the rich, often ambiguous, content of visual artworks to structured semantic representations like RDF triples, several types of failures can occur, leading to a loss or distortion of meaning:

- **Symbolic Overfitting:** This occurs when the ontology or the rules used for mapping visual content to

semantic triples are too closely tailored to a specific, limited set of artworks or a narrow range of iconographic conventions. Such a system might perform well on the data it was designed for but fail to generalize to new, unfamiliar, or variant examples. For instance, an ontology for Christian iconography developed solely from Italian Renaissance art might struggle to accurately represent symbols or themes from Byzantine icons or non-European Christian traditions. This is analogous to the concept of overfitting in machine learning, where a model learns the training data too precisely, including its noise and biases, and consequently performs poorly on unseen data.

- **Metadata Collapse under Visual Plurality:** A single artwork can simultaneously embody multiple meanings, depict various intertwined scenes or sub-narratives, belong to several stylistic categories, or evoke a range of symbolic associations. Forcing this inherent "visual plurality" into a rigid, non-redundant, and often hierarchical metadata structure—as frequently preferred by traditional database logic and some ontological frameworks—can lead to a "collapse." In this collapse, much of the artwork's richness is lost, inadequately represented, or reduced to a single dominant interpretation, thereby obscuring its multifaceted nature. For example, an artwork might be both a historical portrait and a complex allegory; an ontology that forces a choice between these classifications, or only allows for a primary classification, fails to capture its full essence.

The challenge of **Ontological Brittleness versus Evolving Interpretations** further complicates semantic mapping. Ontologies, by their design, aim to create stable, formal, and often canonical representations of knowledge within a specific domain (Prompt 4). However, art historical understanding and interpretation are not static; they are dynamic processes that evolve with new research discoveries, the emergence of new theoretical perspectives, and shifts in cultural values and sensitivities. An ontology developed at a particular point in time, reflecting the scholarly consensus or dominant paradigms of that era, can become "brittle." This means it may be unable to accommodate new interpretations, newly discovered information about an artwork (e.g., a reattribution, a revised dating, an uncovered underdrawing revealing a different initial intent), or emergent conceptual frameworks without requiring significant, costly, and complex revisions. This issue is closely related to "symbolic overfitting": if an ontology is too rigidly defined around current knowledge or a specific interpretive school, it lacks the flexibility to adapt. "Metadata collapse under visual plurality" (Prompt 4) also contributes to this brittleness. If an ontology cannot adequately represent multiple, co-existing, or even conflicting interpretations of an artwork, it fails to capture the dynamic and often contested nature of art historical discourse. The proposed re-framing of the task from "knowledge preservation" to "curation interface" (Prompt 4) directly addresses this. "Knowledge preservation," if interpreted as creating a static snapshot of current understanding, risks fossilizing interpretation. In contrast, a "curation interface" suggests a more dynamic system designed to *support* ongoing scholarly inquiry and curatorial practice. Such an interface would allow the underlying RDF triples to be seen not as definitive truth statements but as one layer of structured data that users can query, augment, debate, and link to alternative interpretations or supporting evidence. Therefore, ontologies for visual heritage should be designed with extensibility, versioning, and mechanisms for representing provenance and disagreement in mind. They should perhaps focus on encoding more fundamental, verifiable descriptive properties (e.g., materials, dimensions, artist, date if known with certainty) while allowing for more fluid, annotative, and attributed layers for interpretive claims, symbolic associations, or contested information. The goal should be to create a system that supports and documents scholarly discourse, rather than one that inadvertently stifles it by prematurely fixing meaning.

4.4. Prompt Drift Variant: "Knowledge Preservation" vs. "Curation Interface"

The reframing of the task from primarily "knowledge preservation" to that of designing a "curation interface" represents a significant conceptual shift with practical implications. If the goal is strictly "knowledge preservation," the emphasis might be on capturing a definitive, stable, and authoritative representation of what is currently "known" about an artwork. This aligns with traditional archival goals of ensuring the long-term persistence of information. However, this perspective can be problematic if "knowledge" in the art historical domain is viewed as fixed, singular, or uncontested, as it often is not.

Conversely, reframing the objective as the creation of a "curation interface" shifts the focus towards developing a system that actively *supports* ongoing human interpretation, scholarly debate, and curatorial activities. In this model, the RDF triples and the underlying ontology might be seen not as immutable truth statements, but as one layer of structured data that users (researchers, curators, students, the public) can query, critically examine, augment with new findings, link to alternative interpretations, and even use to document disagreements or evolving understandings. An effective curation interface would prioritize transparency about the sources and confidence levels of semantic assertions and facilitate the exploration of an artwork's "visual plurality" rather than attempting to reduce it to a single, canonical description. This approach aligns with the idea of an intuitive interface that enables users with diverse backgrounds and expertise to interact meaningfully with archival content.[65]

Section 5: Systemic Failures and Artifact Sensitivity in Complex Computational Heritage Workflows

The integration of multiple computational techniques into complex workflows for cultural heritage documentation and analysis, such as Historic Building Information Modeling (HBIM) and 3D Geographical Information System (GIS) enrichment, introduces new layers of potential failure. These failures often stem from the interaction between different processing stages, the propagation of errors, and the sensitivity of algorithms to artifacts or data characteristics that may not be immediately apparent. This section examines risks associated with semantic compression in HBIM and co-attention failures in 3D GIS enrichment.

5.1. Auto Polygonization in HBIM: Semantic Compression Risk

Historic Building Information Modeling (HBIM) involves the creation of detailed, information-rich 3D models of historic structures, which serve as valuable resources for conservation, restoration planning, research, and public engagement. A common step in HBIM workflows is mesh simplification, or "auto-polygonization," where the geometric complexity of the 3D model (typically represented as a polygon mesh) is reduced to make it more manageable for storage, rendering, or analysis. While simplification is often necessary for practical reasons, it carries a significant risk of erasing or distorting critical architectural features, a phenomenon termed "semantic compression risk."

The analysis of this risk can be approached through several lenses:

- **Semantic detail vs. geometry parsimony:** This represents the core conflict. Mesh simplification algorithms typically prioritize geometric parsimony—that is, reducing the number of polygons while attempting to maintain the overall shape and visual appearance of the model. However, these algorithms often lack an understanding of the *semantic importance* of the features they are simplifying or removing. A small, geometrically simple feature, such as a mason's mark, a subtle decorative element, a specific type of stone joint, or a slight deformation indicating structural stress, might be highly significant from a cultural, historical, or structural engineering perspective. If such a feature is deemed geometrically insignificant by the simplification algorithm, it may be erased or altered, leading to a loss of crucial semantic detail.
- **Cross-resolution bias:** Features that are clearly visible, measurable, and important at high resolution (e.g., in the raw point cloud data or a very dense initial mesh) might be disproportionately lost or distorted when the model is down-sampled or simplified for lower-resolution representations or for different use cases. This bias can lead to inconsistent representations of the heritage asset across different levels of detail, potentially misleading analyses or restoration decisions based on the simplified models.
- **Artifact suppression risk in auto-texturing:** If mesh simplification introduces geometric inaccuracies or removes fine details, subsequent automated texturing processes might attempt to smooth over or visually hide these underlying geometric artifacts. This can result in a model that appears visually plausible but is geometrically false or misleading. The texture might mask the loss of true architectural features, creating a deceptive representation where the surface appearance no longer accurately reflects the underlying form or

material C_H_A_R_A_C_T_E_R_I_S_T_I_C_S.

The prompt to invert this problem into a "loss-aware restoration prompt"—that is, to define what *not* to compress or simplify—points towards a crucial mitigation strategy. This implies the need for semantic segmentation or feature importance mapping to be performed *before* mesh simplification. By identifying and tagging critical architectural features based on their historical, cultural, or structural significance, the simplification algorithm can be guided to preserve these elements with higher fidelity, even if it means a less aggressive overall reduction in geometric complexity.

This highlights the **"Semantic Blind Spot" of Geometrically-Driven Simplification in HBIM.** Mesh simplification algorithms, as commonly employed in HBIM workflows (Prompt 7), are primarily engineered to reduce geometric complexity, typically measured by polygon count, to enhance efficiency in data storage, real-time rendering, and computational analysis. Their optimization criteria are predominantly geometric, focusing on preserving the overall macroscopic shape of the structure and minimizing surface deviation from the original high-resolution model. However, critical architectural features within heritage buildings often possess semantic or cultural significance that is not necessarily proportional to their geometric complexity or scale. A small, intricately carved capital, a specific type of historic brick bond, a subtle wear pattern indicating centuries of use, or a faint inscription might be geometrically minor but historically or culturally paramount. Standard simplification algorithms, lacking inherent semantic awareness, operate with a "semantic blind spot." They may indiscriminately remove, smooth over, or distort these critical features if these features do not contribute significantly to the overall geometric form according to the algorithm's predefined criteria. This is the crux of the "semantic detail vs. geometry parsimony" dilemma. "Cross-resolution bias" and the "artifact suppression risk in auto-texturing" (Prompt 7) are symptomatic of this underlying blind spot. Features that are clearly discernible and significant at one level of resolution can be lost or inaccurately generalized during simplification, and subsequently, textures applied to the simplified mesh might further mask this loss, creating a visually coherent but ultimately misleading representation. Therefore, effective and responsible HBIM workflows necessitate the development and integration of semantically-aware simplification techniques. This could involve pre-annotating features of known importance (requiring expert input), developing algorithms that explicitly weigh geometric simplification against a measure of semantic loss, or utilizing the concept of a "loss-aware restoration prompt" to explicitly guide the simplification process by defining which architectural elements *must* be preserved with high fidelity. Without such semantic guidance, HBIM models risk becoming geometrically efficient but culturally impoverished, failing to serve their full potential as accurate and comprehensive records for heritage preservation and research.

5.2. Co-Attention Failure in 3D GIS Enrichment of Cultural Models

The integration of 3D GIS with cultural heritage models allows for the precise spatial mapping of artifacts, sites, and their contexts. A further step involves "semantic enrichment," where layers of meaning—such as symbolic interpretations, historical data, functional attributions, or intangible associations—are overlaid onto these spatially accurate 3D scans. AI models, often employing co-attention mechanisms, are increasingly used for this task, attempting to link visual or spatial features in the 3D model with relevant semantic labels or descriptions from knowledge bases. However, this process is prone to specific types of failures related to the co-attention mechanism itself.

The failure stack can include:

- **Latent co-attention hallucinations:** Co-attention mechanisms are designed to learn correlations between different modalities (e.g., visual features from a 3D scan and textual concepts from a database). However, in their complex latent spaces, these mechanisms might identify and amplify spurious or coincidental correlations. This can lead to "hallucinated" enrichments, where the AI incorrectly assigns a semantic label or interpretation to a part of the 3D model based on superficial visual or spatial similarities to examples in its training data, even if there is no genuine historical or cultural basis for that association. For instance, a

mundane geological formation within a heritage site might be incorrectly tagged as having ritual significance if it shares some accidental visual features with known ritual sites present in the AI's training set.

- **False spatial-semantic correlation:** The AI model might incorrectly infer a semantic relationship between elements based purely on their spatial proximity, configuration, or geometric properties within the 3D GIS model, without any supporting historical, archaeological, or cultural evidence. For example, it might assume that two artifacts found near each other are symbolically linked or were used together, when their co-location is purely coincidental or due to post-depositional processes.

The lens blend for analyzing these failures highlights critical tensions:

- **GIS precision vs. symbolic inference fragility:** The high geometric precision and apparent objectivity of GIS mapping can create a false sense of confidence in the semantic inferences overlaid upon it. However, symbolic interpretations are often much more fragile, speculative, context-dependent, and open to scholarly debate than the precise spatial data they are attached to. There is a risk that the certainty of the spatial representation lends undue authority to uncertain semantic claims.
- **Annotation collapse under depth-weighted salience:** AI attention mechanisms, if not carefully designed, might be biased towards features that are physically closer to the scanner during data acquisition, larger in size, or more geometrically prominent in the 3D model (i.e., features with high depth-weighted salience). This can lead to an "annotation collapse," where these salient features receive disproportionate attention from the semantic enrichment AI, resulting in their over-interpretation while more subtle, distant, or less geometrically prominent (but potentially equally or more symbolically important) elements are ignored or under-interpreted.

A proposed "recursion layer" involves training an agent to tag false positive enrichments by identifying contradictions against known, curated metadata or established scholarly knowledge. This suggests an iterative refinement process where AI-generated semantic enrichments are continuously validated and corrected, allowing the AI to learn to identify and reduce its own error patterns over time.

This scenario leads to the **"Illusory Correlation" Trap in GIS-Based Semantic Enrichment.** The use of 3D GIS provides highly precise and often visually impressive spatial data for cultural models (Prompt 10). This inherent precision can lend an aura of objectivity and accuracy to the entire dataset, including any semantic layers applied to it. Semantic enrichment layers, frequently driven by AI models employing co-attention mechanisms, attempt to link these precise spatial features with symbolic, historical, or functional meanings (Prompt 10). The "illusory correlation" trap arises when the AI model, through its learning process, identifies statistically significant but semantically spurious connections within its training data or between the visual/spatial features and potential semantic labels. For example, if several known ritual sites in the training dataset happen to be located near a particular type of geological formation or share a common, but unrelated, architectural feature, the AI might erroneously learn to associate that formation or feature with ritual significance in general, even if the observed correlation is purely coincidental or due to sampling bias. This can lead to "latent co-attention hallucinations," where the model "sees" meaningful patterns that are not there, and "false spatial-semantic correlation," where proximity or visual similarity is mistaken for genuine cultural connection. The tension between "GIS precision vs. symbolic inference fragility" (Prompt 10) perfectly captures this predicament: the robustness and exactitude of the underlying spatial data can mask or lend undue credence to the often speculative, probabilistic, or context-dependent nature of the symbolic inferences made by the AI. Furthermore, "annotation collapse under depth-weighted salience" illustrates another facet of this problem, where a purely data-driven attentional mechanism (e.g., focusing on what is physically largest, closest to the sensor, or most geometrically complex) can override a culturally informed understanding of significance, leading to a skewed or incomplete semantic enrichment. Consequently, the semantic enrichment of 3D GIS models for cultural heritage requires robust and continuous validation against external, curated knowledge sources and expert scholarly review, as suggested by the concept of a "recursion layer" for identifying false positives (Prompt 10). The precision of GIS technology should not be mistaken for certainty in symbolic interpretation; human expertise remains crucial to guide, verify, and contextualize AI-driven enrichments to avoid the propagation or creation of new misconceptions based on

these illusory correlations.

Section 6: Towards Critical Frameworks and Ethical Praxis in Computational Visual Heritage

The preceding sections have highlighted numerous challenges and failure points in the application of advanced computational technologies to visual heritage. Synthesizing these observations, this section focuses on the imperative to develop robust evaluation methodologies, establish clear ethical guidelines, and cultivate a critically-aware approach to these powerful tools. The goal is to move beyond uncritical adoption towards a more reflective and responsible engagement that prioritizes the integrity and nuanced understanding of cultural artifacts.

6.1. Defining "Acceptable Loss" and "Fidelity" Beyond Technical Metrics

A recurrent theme throughout the analysis of compression, AI interpretation, and digital pipelines is the profound inadequacy of purely technical metrics for evaluating the impact of computational processes on cultural heritage. Concepts like compression ratios, Peak Signal-to-Noise Ratio (PSNR) for image quality, or BLEU/ROUGE scores for caption generation, while providing some measure of technical performance, often fail to capture the loss of cultural, symbolic, or aesthetic value.

The notion of "acceptable information loss," which is inherent in any lossy compression or dimensionality reduction technique, must be fundamentally redefined in the context of visual heritage. While guidelines exist for acceptable information loss in contexts like web accessibility (e.g., ensuring content is perceivable and operable [66]), these do not readily translate to the semantic and cultural dimensions of heritage objects. The critical question becomes: what constitutes an acceptable loss of symbolic meaning, historical evidence, or aesthetic integrity? The irretrievable data loss associated with lossy compression techniques—which can permanently remove nuance, color, and depth [16]—poses a direct threat if not carefully managed with semantic considerations in mind.

"Cultural fidelity" (as raised in Prompt 1 concerning compressed iconography) and "aesthetic fidelity" (as raised in Prompt 3 concerning color pipelines) need to be operationalized with new frameworks. This is distinct from generic notions of fidelity, such as symbols *of* fidelity (e.g., a dog representing loyalty [68]); here, it refers to the fidelity *of the digital representation to the original artwork's cultural and aesthetic essence*. This may involve developing novel qualitative and quantitative methodologies that assess the preservation of specific symbolic meanings, artistic intent, and culturally relevant visual characteristics. While objective Image Quality Metrics (IQMs) like Multi-Scale Structural Similarity Index (MS-SSIM), Video Multimethod Assessment Fusion (VMAF), Visual Information Fidelity (VIF), and Normalized Laplacian Pyramid (NLPD) have shown correlation with human perception of *image quality* for compressed images [70], their ability to predict or quantify *semantic loss* in complex cultural heritage imagery remains largely an open and critical research question.

This necessitates an **"Evaluative Triad"** for computational heritage processes, moving beyond current paradigms. Often, the evaluation of these processes centers on technical fidelity (e.g., pixel accuracy in reproduction, data completeness in 3D models [71]) or, at best, perceptual fidelity (e.g., the visual similarity of a processed image to the original, as assessed by IQMs like MS-SSIM [70]). However, the analyses across the ten meta-prompts consistently demand consideration of a third, crucial dimension: semantic fidelity. This encompasses the preservation of symbolic meaning, historical accuracy, cultural intent, and interpretive potential (as highlighted in Prompts 1, 2, 3, 4, 5, and others). A critical framework must therefore address this evaluative triad. A computational process might be technically accurate (e.g., a PCA-based compression achieving a low reconstruction error [15]) and even perceptually acceptable (the compressed image still looks "good" to a non-expert observer) but still fail

catastrophically on the semantic dimension if a key symbol is lost, a crucial color relationship is altered, or an important piece of underdrawing is obscured. This requires a decisive move beyond existing metrics. For instance, "failure-informed thresholding" (Prompt 1) is explicitly a semantic evaluation strategy. The historian's critique of AI interpretations (Prompt 2) is a semantic validation. The call to avoid BLEU/ROUGE in favor of assessing symbolic misalignment (Prompt 5) is a direct demand for semantic metrics. Therefore, future research in computational heritage must prioritize the development and rigorous validation of new metrics and methodologies that explicitly assess semantic fidelity. This will likely involve hybrid approaches that combine advanced computational analysis with structured expert evaluation protocols [65], and the creation of tools that can trace the impact of various processing steps on semantically salient features within an artwork.

6.2. The Role of Expert Critique and Human-in-the-Loop Validation

The limitations of purely automated assessment underscore the indispensable role of human expertise in validating and critiquing the outputs of computational processes applied to visual heritage. The meta-prompts repeatedly emphasize this necessity, for example, through the call for a historian's critique of AI-generated symbolic interpretations (Prompt 2), the concept of failure-informed thresholding for compression requiring semantic judgment (Prompt 1), and the proposal of a recursion layer for validating 3D GIS enrichments against known metadata (Prompt 10).

Comparisons between human and machine image analysis reveal complementary strengths: humans often excel at nuanced classification and pattern recognition based on learned "key features" (KFs) that are culturally and contextually informed, while machines typically excel at precise quantification and processing vast datasets.[23] For complex visual data like iconography, human expertise in recognizing culturally-specific KFs and interpreting their symbolic significance remains paramount. Therefore, combining computational metrics with structured expert assessment is crucial for a holistic evaluation of outcomes, particularly for detecting subtle semantic loss or misinterpretation.[65]

Explainable AI (XAI) aims to make the decision-making processes of AI models more transparent and understandable to humans.[44] However, XAI methods themselves, including gradient-based techniques like Grad-CAM, are not infallible and require critical evaluation regarding their faithfulness, robustness, and potential to mislead.[43] An explanation that appears plausible might not accurately reflect the model's true reasoning or could even mask underlying biases.

6.3. Ethical Frameworks for Computational Interventions in Sacred and Cultural Art

The application of advanced computational technologies—such as aggressive data compression, AI-driven interpretation, or algorithmic alteration—to sacred art and culturally sensitive visual heritage raises profound ethical questions that go beyond general data ethics.[75] Sacred art, for instance, often derives its value not merely from its aesthetic qualities but from its precise symbolism, its role in ritual practice, and its embodiment of spiritual truth.[75] Any computational intervention that alters these core attributes, misinterprets sacred meanings, or decontextualizes the artwork can be seen as a form of violation, potentially causing offense or spiritual harm.

Broader ethical considerations in the digital cultural heritage domain include ensuring data accuracy, responsible interpretation, preventing misuse of cultural data, respecting data sovereignty (particularly for Indigenous heritage), and actively working to avoid the perpetuation or amplification of biases and cultural distortions through computational means.[77] The "aesthetic inversion" that might result from flawed HDR tone mapping (Prompt 6) or the creation of "semantic voids" by AI emotion recognition systems applied to stylized art (Prompt 8) can be

considered ethical failures if they lead to the misrepresentation, trivialization, or de-sacralization of cultural heritage. Developing ethical frameworks specific to computational visual heritage requires careful consideration of these unique sensitivities.

6.4. Addressing Systemic Biases and Ensuring Interpretive Plurality

Computational tools and AI models are not neutral; they can inherit and even amplify biases present in their training data, algorithmic design, or the assumptions of their creators. Examples include potential color biases embedded in imaging tools (Prompt 3) or the drift and amplification of biases in AI-driven symbol interpretation, particularly concerning mythological, religious, or culturally specific symbols (Prompt 5[47]).

Robust frameworks must be developed to systematically detect, measure, and mitigate these biases. This involves not only curating more diverse and representative training datasets for AI models but also designing algorithms that are inherently more sensitive to cultural context and less prone to simplistic generalizations. Furthermore, the goal of ensuring "interpretive freedom" (Prompt 4) in the face of structured data requirements (such as those imposed by RDF ontologies) is critical. Systems for representing visual heritage knowledge should be designed to accommodate multiple, potentially conflicting, interpretations rather than imposing a single, monolithic "correct" view. This allows for the representation of scholarly debate, evolving understandings, and the inherent ambiguities often present in cultural artifacts.

The development and application of computational methods in visual heritage must be guided by **"Algorithmic Humility" as an ethical imperative.** Many of the analyzed prompts highlight significant AI failures: semantic drift in VLPs (Prompt 2), pervasive hallucinations in captioning (Prompt 5), misclassification of emotions in symbolic art (Prompt 8), and co-attention hallucinations in 3D GIS enrichment (Prompt 10). These failures underscore the current limitations of AI in comprehensively understanding the profound complexities of human culture, symbolism, and artistic expression.[24] While existing ethical guidelines for AI often focus on crucial issues like bias, privacy, and misuse [77], a deeper ethical concern in the computational heritage domain is the risk of "epistemic arrogance." This occurs when algorithmic outputs are presented, perceived, or accepted as objective, unassailable truth, thereby overriding nuanced humanistic interpretation, marginalizing alternative scholarly perspectives, or simplifying complex cultural narratives. The call for "epistemological caution" (as a guiding tone for this report) and the inclusion of reflexive loops and critical self-assessment mechanisms in the prompts (e.g., Prompts 1, 3, 6) strongly suggest a need for "algorithmic humility." This entails a conscious and explicit acknowledgment of the inherent limitations, potential biases, and uncertainties associated with computational systems when they are applied to the rich and often ambiguous domain of cultural heritage. Practically, this involves transparently communicating the uncertainty levels of algorithmic outputs, flagging potential errors or areas of low confidence, designing systems that facilitate robust human oversight and correction (as exemplified by the recursion layer in Prompt 10), and prioritizing the preservation of interpretive plurality (Prompt 4) over computational neatness or the imposition of a single, algorithmically derived "truth." The profound responsibility involved in handling sacred or culturally paramount art, as underscored by Schuon's principles [75], further amplifies this ethical demand. Therefore, ethical praxis in computational visual heritage must extend beyond mere technical safeguards to actively cultivate a research and development culture characterized by critical self-reflection and profound humility regarding the capabilities and outputs of AI. This means designing systems that are not just "explainable" in a technical XAI sense but are also capable of explicitly articulating their own limitations and potential for error when operating in culturally sensitive and interpretively complex domains.

Conclusion and Future Imperatives

This critical examination of advanced imaging and AI in visual heritage has revealed a series of fundamental tensions: between computational power and semantic integrity; between technical fidelity and cultural or aesthetic faithfulness; and between the drive for automation and the irreplaceable role of expert human judgment. The

diverse "failure points" identified across the ten analytical prompts are not to be dismissed as mere technical bugs or isolated shortcomings. Instead, they should be recognized as crucial sites for critical inquiry and methodological innovation. These failures illuminate the underlying assumptions, inherent limitations, and potential biases of current computational paradigms when they are applied to the nuanced, polysemic, and culturally rich domain of visual heritage.

The path forward requires a paradigm shift from uncritical adoption of new technologies to a more reflective, responsible, and ethically grounded approach. Several key imperatives emerge:

1. **Development of New, Hybrid Evaluation Metrics:** There is an urgent need for evaluation metrics that transcend purely technical or perceptual assessments. These new metrics must integrate technical, perceptual, and, crucially, semantic criteria to provide a holistic understanding of a technology's impact on cultural heritage. This involves operationalizing concepts like "cultural fidelity" and "semantic loss."
2. **Prioritization of "Loss-Aware" and "Semantic-Aware" Processing:** Computational techniques applied to visual heritage should be designed with an explicit awareness of potential information loss, particularly semantic loss. Efficiency-driven approaches must be balanced with, or superseded by, methods that actively seek to preserve symbolic meaning, artistic intent, and cultural context.
3. **Mandating Human-in-the-Loop Validation and Expert Critique:** Given the current limitations of AI in nuanced interpretation, human expertise remains indispensable. Workflows involving AI-driven analysis or interpretation of cultural heritage must incorporate robust mechanisms for validation, critique, and correction by domain experts (art historians, conservators, archaeologists, cultural custodians).
4. **Investing in Robust and Faithful Explainable AI (XAI) for Cultural Heritage:** Research into XAI methods tailored for cultural heritage applications is vital. The focus should be on developing techniques that offer true faithfulness to the model's reasoning, are robust to perturbation, and, importantly, are capable of exposing rather than masking model limitations, biases, and uncertainties when dealing with complex visual material.
5. **Fostering Deep Interdisciplinary Collaboration from Inception:** The development of computational tools for visual heritage should involve close collaboration between technologists, humanities scholars, and heritage professionals from the very outset of design and conceptualization, not merely in the final application or testing phase. This ensures that cultural and interpretive requirements inform technological development.
6. **Developing Specific Ethical Guidelines Emphasizing "Algorithmic Humility":** Beyond general AI ethics, specific guidelines are needed for the computational treatment of visual heritage. These must address the unique challenges of representing, interpreting, and potentially altering sacred, sensitive, or culturally significant visual material. A core principle should be "algorithmic humility"—a commitment to transparency about limitations, the explicit acknowledgment of uncertainty, and the active preservation of interpretive plurality.

The future of computational visual heritage lies not in the pursuit of fully automated systems that aim to replace human expertise, but in the development of critically-aware, ethically-grounded, and collaboratively designed tools. Such tools should augment our collective capacity to understand, preserve, interpret, and engage with the profound richness and diversity of the human past as embodied in its visual legacies. The algorithmic gaze must be guided by human wisdom, cultural sensitivity, and an unwavering respect for the integrity of the heritage it seeks to illuminate.

Appendix A

Table 2: Computational Techniques, Heritage Applications, and Critical Failure Lenses (Summary)

Meta-Prompt Ref.	Core Technology/Method	Primary Application Area in Visual Heritage	Key Failure Lenses / Critical Questions

Prompt 1	Principal Component Analysis (PCA)	Compression of UV-Fluorescence Iconography	Signal degradation vs. cultural fidelity; Latent symbolism retrieval error; Machine-readable vs. human-interpretable.
Prompt 2	Vision-Language Models (VLPs) (e.g., BLIP, Flamingo)	Symbolic Meaning Inference in Paintings	Scene consistency vs. accurate symbolic attribution; Semantic drift; Visual grounding vs. textual heritage.
Prompt 3	YRGB vs. Lab Color Transformations	Digitized Color Pipelines for Artworks	Technical fidelity vs. aesthetic fidelity; Perceived chroma mistranslation; Color biases in tools.
Prompt 4	RDF Triples, Ontology Modeling	Semantic Archiving of Digitized Art	Cognitive load vs. interpretative freedom; Compression of visual nuance into triples; Database logic vs. visual ambiguity.
Prompt 5	Grad-CAM++ Guided Captioning, Co-attention VLMs	Iconographic Caption Validation for Artworks	Symbolic misalignment; Artifact sensitivity; Visual-text co-entanglement; Explainability flaws; Bias drift (mythological vs. secular); Learned iconography vs. style mimicry.
Prompt 6	Adaptive Logarithmic HDR Systems, Layer Masks	Tone Mapping for Artworks	Computational tone accuracy vs. perceived depth collapse; Layer mask opacity ambiguity as hallucination vector; System detecting its own "aesthetic inversion."
Prompt 7	Auto Polygonization (Mesh Simplification) in HBIM	HBIM Workflows for Restoration Readiness	Semantic detail vs. geometry parsimony; Cross-resolution bias; Artifact suppression risk in auto-texturing.
Prompt 8	CNN Texture Maps for Emotion Recognition	Emotion Inference from Stylized/Symbolic Artworks	Misclassification of ritual scenes; Stylization-induced semantic void; Texture dominance erasing emotional intention.
Prompt 9	Optical Flow in Aerial Photogrammetry	HDR Footage for Heritage Site Reconstruction	Dynamic range exceeding optical flow compensation; Information degradation under high luminance contrast; SfM vs. depth + tone fusion.
Prompt 10	Co-Attention in 3D GIS Enrichment	Semantic Enrichment of 3D Cultural Scans	Latent co-attention hallucinations; False

			spatial-semantic correlation; GIS precision vs. symbolic inference fragility; Annotation collapse under depth-weighted salience.

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